

TOAST Stroke Classification

Trial of Org 10172 in Acute Stroke Treatment (TOAST) diagnostic criteria for ischemic stroke etiology.

1. Large-Artery Atherosclerosis (LAA)

- **Clinical:** Cortical signs (aphasia, neglect, gaze deviation) or brainstem/cerebellar syndrome.
- **Imaging:** Cortical or subcortical/cerebellar/brainstem infarct matching the vascular territory.
- **Vascular:** > 50% stenosis or occlusion of the relevant major extracranial (carotid, vertebral) or intracranial artery.
- **Exclusion:** Must exclude a high-risk cardioembolic source.

2. Small-Vessel Occlusion (SVO / Lacune)

- **Clinical:** Classic lacunar syndrome (pure motor, pure sensory, sensorimotor, ataxic hemiparesis, clumsy hand) **WITHOUT** cortical signs.
- **Imaging:** Normal scan or subcortical/brainstem lesion < 1.5 cm in diameter.
- **Vascular/Cardiac:** Relevant artery must lack >50% stenosis, and patient must lack high-risk cardioembolic sources.

3. Cardioembolism (CE)

Illustrative historical TOAST categories (1993), not a current anticoagulation checklist.

HIGH-RISK SOURCES:

- Mechanical prosthetic valve
- Mitral stenosis w/ AFib
- Atrial fibrillation (other than lone AF)
- Left atrial / LAA thrombus
- Sick sinus syndrome
- Recent MI (< 4 weeks)
- Dilated cardiomyopathy or LV thrombus
- Infective endocarditis

MEDIUM-RISK SOURCES:

- Mitral valve prolapse
- Mitral ring calcification
- Mitral stenosis w/o AFib
- Left atrial turbulence/smoke
- PFO or atrial septal aneurysm
- Lone atrial fibrillation (historical term)
- Atrial flutter (isolated)
- Bioprosthetic heart valve
- Nonbacterial endocarditis

4. Other Determined Etiology (ODE)

- **Clinical/Imaging:** Infarction of any size with diagnostic proof of a rare/specific underlying mechanism:
- Arterial dissection (e.g. carotid or vertebral dissection)
- CNS vasculitis or systemic vasculopathy
- RCVS (Reversible Cerebral Vasoconstriction Syndrome)
- Moya-Moya disease, CADASIL, or Fibromuscular Dysplasia
- Prothrombotic/hypercoagulable state (APLS, cancer, DIC)

5. Undetermined Etiology (UDE)

- **Two or more potential causes:** e.g. 60% carotid stenosis AND atrial fibrillation (unable to assign single primary cause).
- **Incomplete evaluation:** Imaging or cardiac workup pending/incomplete.
- **Cryptogenic:** no cause identified after an appropriate evaluation. ESUS is a narrower nonlacunar-infarct category with defined vascular, cardiac and other-cause exclusions; the terms are not interchangeable.

Etiologic Evaluation: Tailor to the Clinical Question

- ✓ **Parenchymal:** MRI Brain (DWI/ADC) preferred, or CT Head.
- ✓ **Vascular:** CTA or MRA Head & Neck (or Carotid Duplex + TCD).
- ✓ **Rhythm:** EKG + Continuous Telemetry ≥ 24h (or loop recorder).
- ✓ **Cardiac:** select TTE, TEE and shunt testing according to the suspected mechanism and whether results would change management.

WHY SUBTYPE MATTERS — 2026 SECONDARY PREVENTION

Prevention depends on the **specific mechanism**. AF-related stroke often warrants anticoagulation; infective endocarditis and prosthetic valves require distinct management and drug selection. Noncardioembolic stroke generally uses antiplatelet therapy, with short-course DAPT only in eligible settings. ESUS alone is not an indication for empiric DOAC therapy.

The factor Xla inhibitor class (new for 2026): these oral agents aim to uncouple antithrombotic efficacy from bleeding risk. Both phase-2 stroke trials were neutral — **PACIFIC-Stroke** (asundexian) and **AXIOMATIC-SSP** (milvexian, a covert-infarct-heavy composite). **OCEANIC-STROKE** (phase 3) was then the first positive trial of the class: adding oral **asundexian** to antiplatelet therapy in non-cardioembolic stroke/high-risk TIA cut recurrent ischemic stroke (HR 0.74) with no significant excess of major bleeding. The milvexian phase-3 **LIBREXIA-STROKE** is ongoing. Note the class does NOT replace guideline anticoagulation for AF (OCEANIC-AF stopped early for inferiority to apixaban).

Original Study: Adams HP Jr, et al. TOAST. *Stroke*. 1993;24:35-41. **PMID: 7678184** (<https://pubmed.ncbi.nlm.nih.gov/7678184/>).

AHA/ASA Guideline: Kleindorfer DO, et al. 2021 Stroke Prevention. *Stroke*. 2021;52:e364-e467. **PMID: 34024117** (<https://pubmed.ncbi.nlm.nih.gov/34024117/>).

Factor Xla class: OCEANIC-STROKE (Sharma M, et al. *N Engl J Med*. 2026;394:1467-1479. **PMID: 41985132** (<https://pubmed.ncbi.nlm.nih.gov/41985132/>)) • PACIFIC-Stroke (Shoamanesh A, et al. *Lancet*. 2022;400:997-1007. **PMID: 36063821** (<https://pubmed.ncbi.nlm.nih.gov/36063821/>)) • AXIOMATIC-SSP (Sharma M, et al. *Lancet Neurol*. 2024;23:46-59. **PMID: 38101902** (<https://pubmed.ncbi.nlm.nih.gov/38101902/>)).